PROFESSIONAL VALUES AND ETHICS

Cybercrime and Engineer's Ethical Role

CASE STUDY

Submitted by:

Huzaifa Shahbaz
21-CE-009

Aqsa Shah
21-CE-039

Submitted to:

MAAM RUQYA ZUBAIR

Department of Computer Engineering,

HITEC University, Taxila

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ABSTRACT:

The rapid advancement of digital technology transformed modern society and made it so connected and innovative as it has ever been. However, along with that rapid transformation was an increase in cybercrime and ethical challenges for engineers to design, maintain, and secure digital systems. This study unfolds the ethical responsibility of the engineer in fighting cybercrime within the scope of Pakistan Engineering Council (PEC) Code of Ethics. It includes scenarios such as data breach, hacking, intellectual property theft, and misuse of personal data, analyzed under professional ethics. It emphasizes the most important roles of engineers in conforming to ethical standards in ensuring the security and integrity of systems and protection of the public interest. In addition, it adds the PEC guiding principles part, where coming up with the importance of public safety over profession integrity, and never getting into conflict of interest would guide the engineer in ethical dilemmas posed as a result of cybercrime. This has led engineers to include ethical decision-making in their practice so that the cybercrime risks ensuring public trust in technology systems are minimized.

INTRODUCTION:

The finding is that increased illegal activities from accessing phishing and data breach are all among the growing threats that have come throughout the world during the digital modern age. The extent to which using the computer engineers affects trends in this area would result in obsolescence of the risk. Engineers therefore build secure systems while maintaining their ethical responsibility. The principles of integrity, accountability, and public welfare as mentioned in the Pakistan Engineering Council (PEC) Code of Ethics intend to guide engineers to prioritize cybersecurity and report faults. By following the guidelines, engineers would be the first pre-emptive line against misuse of information and security among members of the public. This case study examines the ethical role of engineers in terms of addressing cybercrime - addressing the principles defined in PEC and giving several examples in reality. Innovation is thus meant to go hand in hand with responsibility since these two are necessary in ensuring technology systems that are safe and trustworthy.

OBJECTIVES:

1: Inquire into the forms and nature of cybercrimes prevalent in society: breaches of data, hacking, identity theft, and misappropriation of software systems.

2: Pinpoint the role of computer engineers in the development of secure systems that will be prepared to respond to cybercrime while adhering to the ethical and professional standards of behavior.

3: Consider the Code of Ethics established by the Pakistan Engineering Council in reference to the responsibilities of engineers to ensure public safety, confidentiality, and professional integrity in cyberspace.

4: Establish a guideline for engineers on making ethical decisions in dilemmas such as compromises to cyber security or improper use of their technological designs.

5: Compare the contributions of ethical and unethical engineering practice to the industry-based effect such as example case studies of computer engineers.

6: Study the role of engineers in creating awareness about cyber-crimes and the obligation to report and prevent unethical practices as prescribed by PEC.

7: Propose recommendations for engineers on how they can implement cyberspace ethics in software development, system architecture, and data protection initiatives.

HISTORY OF CYBERCRIME:

Cybercrime exists in the realm of old-timey origins, which have their bases as early as the 60s to the 80s and were catalyzed by the advent of mainframe computers and telephone phreaking. In the 1980s, computer viruses and malware were invented, starting with the Brain virus. The emerging threats of the Internet in the 1990s were hacking, email phishing, and data breaches. Since the 2000s, types of cybercrime such as ransomware, DoS (denial of service) attack, identity theft, and advanced persistent threat have emerged. Among great milestones are the first-ever online banking crime case in 1994, the Love Bug Virus in 2000, and the 2017 WannaCry ransomware attack.

ENGINEER'S ETHICAL ROLE IN CYBERCRIME MITIGATION:

The PEC Code of Ethics describes that professional ethics are components of computer engineering ethics. It further stresses the ethical system for data protection and vulnerability. Furthermore, engineers must adhere to it, dealing with privacy versus security, government surveillance, and hacking tools. They will also have ethical dilemmas. The PEC Code of Ethics applies justice, honesty, and accountability and contributes to serving the public interest on a fair and impartial basis, with strong cybersecurity and avoiding conflicts of interest.

PEC CODE OF ETHICS:

The Prevention of Electronic Crimes Act 2016 assigned priority to the engineers to promote the safety, health, and welfare of people by protecting critical systems from cyber-attacks. They should keep away from deceiving acts in cybersecurity and respect confidentiality. Engineers in the line of cybersecurity equal failure to ensure compliance with data protection laws such as GDPR or Pakistan PECA and develop systems with highly authentic credentials and excellent encryption to prevent unauthorized access. Like every new invention or scheme, the contraption here also demanded careful and stringent examination of the essential provisions and requirements each time. Indeed, the Prevention of Electronic Crimes Act contains a lot of things, considering protection to be a ten-speed vehicle that can run at different speeds: the setting of a culture in which people can start refuting cybercrimes.

RELEVANT PEC CODE OF CONDUCT ARTICLE:

Article no 1: Honesty and Integrity

Engineers must act with honesty and integrity in all professional matters, avoiding any conduct that might harm individuals or society.

Article no 2:Competency

An engineer should only perform tasks that he/she is trained and competent to do and should never put his or her work into use for unethical acts such as cyber-crimes.

Article no 3: Public Interest

An engineer must have a public interest to counteract the threat of their work to society. Both to their advantage and the frontiers of disadvantage, be it life or an aspect thereof.

Article 5-Confidentiality:

It prohibits engineers from the unauthorized retrieval of sensitive and confidential information as it is more imperative in preventing cybercrimes.

Article 7-Social Responsibility:

It prohibits engineers from the unauthorized retrieval of sensitive and confidential information as it is more imperative in preventing cybercrimes.

CONCLUSION:

In conclusion, engineering ethics helps the engineers most in developing skills to tackle cybercrime with the guidance of the PEC Code of Ethics by the Pakistan Engineering Council. They are sincere and honest, keep above all things the public interest while making any innovation, maintain confidentiality and social responsibility. Engineering makes secure systems, compliance with specific laws such as PECA, and endorsement of ethical practices serve to provide public trust in technology. Ethical decision-making and accountability of an engineer become imperative in preventing the risks associated with cybercrime but still encourage innovation with responsibility. The engineers, through awareness, prevention, and ethical conduct, contribute immensely toward the reduction of issues in cybersecurity and a safer digital environment.